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Antenna module and communication device mounted with same

Abstract

An antenna module includes a dielectric substrate that has an upper surface and a lower surface; radiating elements that have a flat-plate shape; and dielectric layers. The radiating elements are arranged in the dielectric substrate and can radiate respective radio waves in mutually-different frequency bands. The dielectric layer is arranged in a manner to cover a first region in which the radiating element is arranged. The dielectric layer is arranged in a manner to cover a second region in which the radiating element is arranged. Dielectric constants of the dielectric layers are higher than a dielectric constant of the dielectric substrate. The first region and the second region are adjacent to each other. The radiating element overlaps with the radiating element or the radiating element in plan view in a normal direction of the dielectric.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of PCT/JP2022/029278, filed on Jul. 29, 2022, designating the United States of America, which is based on and claims priority to Japanese Patent Application No. JP 2021-154353 filed on Sep. 22, 2021. The entire contents of the above-identified applications, including the specifications, drawings and claims, are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to an antenna module and a communication device mounted with

the same, and more specifically relates to a technique for realizing broadband in antenna characteristics.

BACKGROUND ART

(2) Japanese Unexamined Patent Application Publication No. 2003-198230 (Patent Document 1) discloses the configuration for improving antenna characteristics by using dielectrics, whose thickness differ depending on corresponding frequencies, for respective antenna portions in an antenna in which a plurality of antenna portions corresponding to mutually-different respective frequency bands are arranged on the same substrate.

(3) International Publication No. 2020/066453 (Patent Document 2) discloses the configuration in which an upper portion of a dielectric member, which is arranged on a patch antenna, is sealed with sealing resin in an antenna device including the patch antenna having a flat-plate shape.

(4) U.S. patent Ser. No. 10/784,593 (Patent Document 3) discloses the configuration in which two dielectric layers are arranged on a patch antenna on a lower frequency side and one dielectric layer is arranged on a patch antenna on a higher frequency side in a dual-band type patch antenna.

CITATION LIST

Patent Documents

(5) Patent Document 1: Japanese Unexamined Patent Application Publication No. 2003-198230

Patent Document 2: International Publication No. 2020/066453 Patent Document 3: U.S. patent Ser. No. 10/784,593

SUMMARY OF DISCLOSURE

Technical Problem

(6) In recent years, development of communication devices compatible with a plurality of communication standards has been progressing. Such communication devices are required to transmit and receive radio waves in mutually-different frequency bands defined by respective communication standards, and antennas corresponding to three or more frequency bands are sometimes arranged on the same dielectric substrate. A wide frequency bandwidth is desired for each of the frequency bands.

(7) In general, parameters suitable for antenna characteristics (such as a dielectric constant) differ depending on a target frequency band. In the configuration in which antennas corresponding to three or more mutually-different frequency bands are arranged on the same substrate as described above, it may be impossible to optimize parameters for all of the antennas.

(8) The present disclosure has been made to solve the above-mentioned problems and an object of the present disclosure is to realize broadband in antenna characteristics of each radiating element in an antenna module in which radiating elements corresponding to three or more mutually-different frequency bands are arranged.

Solution to Problem

(9) An antenna module according to a first aspect of the present disclosure includes a dielectric substrate that has a first surface and a second surface, first to third radiating elements that have a flat-plate shape, a first dielectric layer, and a second dielectric layer. The first to third radiating elements are arranged in the dielectric substrate and can radiate respective radio waves in mutually-different frequency bands. The first dielectric layer is arranged on the first surface of the dielectric substrate in a manner to cover a first region in which the first radiating element is arranged. The second dielectric layer is arranged on the first surface of the dielectric substrate in a manner to cover a second region in which the third radiating element is arranged. A dielectric constant of the first dielectric layer and a dielectric constant of the second dielectric layer are higher than a dielectric constant of the dielectric substrate. The first region and the second region are adjacent to each other. The first radiating element can radiate a radio wave in a first frequency band. The second radiating element can radiate a radio wave in a second frequency band that is higher than the first frequency band. The third radiating element can radiate a radio wave in a third frequency band that is higher than the second frequency band. The second radiating element overlaps with the

first radiating element or the third radiating element in plan view in a normal direction of the dielectric substrate.

(10) An antenna module according to a second aspect of the present disclosure includes a dielectric substrate that has a first surface and a second surface, first to third radiating elements that are arranged in the dielectric substrate, a first dielectric layer, and a second dielectric layer. Each of the first to third radiating elements includes a plurality of electrodes having a flat-plate shape. The first dielectric layer is arranged on the first surface of the dielectric substrate in a manner to cover a first region in which the electrodes of the first radiating element are arranged. The second dielectric layer is arranged on the first surface of the dielectric substrate in a manner to cover a second region in which the electrodes of the third radiating element are arranged. The first to third radiating elements can radiate radio waves in mutually-different frequency bands. A dielectric constant of the first dielectric layer and a dielectric constant of the second dielectric layer are higher than a dielectric constant of the dielectric substrate. The first region and the second region are adjacent to each other. The first radiating element can radiate a radio wave in a first frequency band. The second radiating element can radiate a radio wave in a second frequency band that is higher than the first frequency band. The third radiating element can radiate a radio wave in a third frequency band that is higher than the second frequency band. The electrodes of the second radiating element overlap with the electrodes of the third radiating element in plan view in a normal direction of the dielectric substrate.

Advantageous Effects of Disclosure

(11) In the antenna module according to the present disclosure, three types of radiating elements, which can radiate respective radio waves in mutually-different frequency bands, are provided in the same dielectric substrate, and each of two radiating elements of the three are arranged in the first region and the second region that are adjacent to each other. The rest of the radiating elements forms a stack structure with one of the other radiating elements, and mutually-different respective dielectric layers are arranged to cover the first region and the second region. In such a configuration, by individually setting the dielectric layers suitable for the radiating element arranged in the first region and the radiating element arranged in the second region, broadband in antenna characteristics of each radiating element can be realized.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram of a communication device to which an antenna module according to a first embodiment is applied.

(2) FIG. 2 is a plan view and a side perspective view of the antenna module of FIG. 1.

(3) FIG. 3 is a diagram for explaining an advantageous effect of a dielectric layer in antenna characteristics of radiating elements forming a stack structure, in the antenna module of FIG. 1.

(4) FIG. 4 is a side perspective view of an antenna module according to a first modification.

(5) FIG. 5 is a side perspective view of an antenna module according to a second modification.

(6) FIG. 6 is a plan view of an antenna module according to a second embodiment.

(7) FIG. 7 is a side perspective view of an antenna module according to a third modification.

(8) FIG. 8 is a side perspective view of an antenna module according to a fourth modification.

(9) FIG. 9 is a side perspective view of an antenna module according to a fifth modification.

(10) FIG. 10 is a side perspective view of an antenna module according to a sixth modification.

(11) FIG. 11 is a plan view of an antenna module according to a seventh modification.

(12) FIG. 12 is a plan view of an antenna module according to a third embodiment.

DESCRIPTION OF EMBODIMENTS

(13) Embodiments according to the present disclosure will be described in detail below with

reference to the accompanying drawings. Here, the same reference characters are given to the same or corresponding portions and the description thereof will not be repeated.

First Embodiment

(14) (Basic Configuration of Communication Device)

(15) FIG. 1 is an example of a block diagram of a communication device **10** to which an antenna module **100** according to this first embodiment is applied. The communication device **10** is, for example, a portable terminal such as a cellular phone, a smartphone, and a tablet, or a personal computer with a communication function. An example of a frequency band of a radio wave used for the antenna module **100** according to the present embodiment is a millimeter wave band with a center frequency of 28 GHz, 39 GHz, or 60 GHz, for example, but radio waves in frequency bands other than the above are also applicable.

(16) Referring to FIG. 1, the communication device **10** includes the antenna module **100** and a BBIC **200**, which constitutes a baseband signal processing circuit. The antenna module **100** includes an RFIC **110**, which is an example of a feed device, and an antenna device **120**. The communication device **10** up-converts a signal, which is transmitted from the BBIC **200** to the antenna module **100**, to a radio frequency signal at the RFIC **110** and radiates the radio frequency signal from the antenna device **120**. Further, the communication device **10** transmits a radio frequency signal, which is received at the antenna device **120**, to the RFIC **110** so as to down-convert the radio frequency signal and process the obtained signal at the BBIC **200**.

(17) The antenna module **100** is a so-called multi-band type antenna module that can radiate radio waves in three mutually-different frequency bands. The antenna device **120** includes a plurality of radiating elements **121** that radiate radio waves in a first frequency band f1, a plurality of radiating elements **122** that radiate radio waves in a second frequency band f2, and a plurality of radiating elements **123** that radiate radio waves in a third frequency band f3. In the antenna module **100**, the first frequency band f1 of the radiating element **121** is relatively the lowest frequency and the third frequency band f3 of the radiating element **123** is relatively the highest frequency. That is, the magnitude relationship among the frequencies of radio waves radiated from the three radiating elements is expressed as $f1 < f2 < f3$.

(18) In the example illustrated in FIG. 1, the radiating element **121** and the radiating element **122** are arranged in a stacked manner in a dielectric substrate, and the radiating element **123** is arranged in the dielectric substrate. Pairs of the radiating element **121** and the radiating element **122** which are stacked and the radiating elements **123** are alternately arranged in one direction. FIG. 1 illustrates the example of the configuration of the antenna device **120** in which the radiating elements are arranged in a two-dimensional array, but the antenna device **120** may have a configuration in which radiating elements are aligned in a one-dimensional array. Alternatively, the antenna device **120** may have a configuration in which the radiating elements **121**, **122**, and **123** are provided one by one. In the present embodiment, each of the radiating elements **121**, **122**, and **123** is a patch antenna having a flat-plate shape.

(19) The RFIC **110** includes a feed circuit corresponding to each frequency band, in other words, a feed circuit corresponding to each radiating element. Namely, the RFIC **110** includes a feed circuit **110A** corresponding to the radiating elements **121**, a feed circuit **110B** corresponding to the radiating elements **122**, a feed circuit **110C** corresponding to the radiating elements **123**. FIG. 1 only illustrates the configuration corresponding to four radiating elements for the feed circuit **110A** corresponding to the radiating elements **121** and omits the illustration of the same configurations corresponding to other radiating elements, for the sake of simpler description. That is, the configurations of the feed circuits **110B** and **110C** are the same as the configuration of the feed circuit **110A**.

(20) The feed circuit **110A** includes switches **111A** to **111D**, **113A** to **113D**, and **117**; power amplifiers **112AT** to **112DT**; low noise amplifiers **112AR** to **112DR**; attenuators **114A** to **114D**; phase shifters **115A** to **115D**; a signal synthesizer/demultiplexer **116**; a mixer **118**; and an

amplifying circuit **119**.

(21) In transmitting a radio frequency signal, the switches **111A** to **111D** and **113A** to **113D** are switched to the power amplifiers **112AT** to **112DT** sides and the switch **117** is connected to a transmission amplifier of the amplifying circuit **119**. In receiving a radio frequency signal, the switches **111A** to **111D** and **113A** to **113D** are switched to the low noise amplifiers **112AR** to **112DR** sides and the switch **117** is connected to a reception amplifier of the amplifying circuit **119**.

(22) A signal transmitted from the BBIC **200** is amplified in the amplifying circuit **119** and up-converted in the mixer **118**. A transmission signal that is the up-converted radio frequency signal is demultiplexed into four signals in the signal synthesizer/demultiplexer **116** and fed to respective mutually-different radiating elements **121** through four respective signal paths. At this time, the directivity of the antenna device **120** can be adjusted by individually adjusting phase levels of the phase shifters **115A** to **115D** arranged on respective signal paths. The attenuators **114A** to **114D** adjust strength of a transmission signal.

(23) Reception signals which are radio frequency signals received by respective radiating elements **121** pass through four respective different signal paths and synthesized in the signal synthesizer/demultiplexer **116**. The synthesized reception signal is down-converted in the mixer **118** and amplified in the amplifying circuit **119** to be transmitted to the BBIC **200**.

(24) The RFIC **110** is formed as one chip of integrated circuit component having the above-described circuit configuration, for example. Alternatively, the RFIC **110** may be formed as an individual integrated circuit component for each feed circuit. Further, devices (switch, power amplifier, low noise amplifier, attenuator, phase shifter) corresponding to each radiating element may be formed as one chip of integrated circuit component for each corresponding radiating element.

(25) (Configuration of Antenna Module)

(26) The configuration of the antenna module **100** according to the first embodiment will now be described in detail with reference to FIG. 2. FIG. 2 is a diagram illustrating the antenna module **100** according to the first embodiment. In FIG. 2, a plan view (FIG. 2(A)) of the antenna module **100** is illustrated on the upper side, and a side perspective view (FIG. 2(B)) is illustrated on the lower side. Here, FIG. 2 illustrates an example of the configuration in which each of the radiating elements **121**, **122**, and **123** is composed of one electrode, for the sake of simpler description.

(27) The antenna module **100** includes a dielectric substrate **130**, feed wiring **141**, feed wiring **142**, feed wiring **143**, dielectric layers **151** and **152**, and a ground electrode GND in addition to the radiating elements **121**, **122**, and **123** and the RFIC **110**. In the following description, a normal direction of the dielectric substrate **130** (a direction of radio wave radiation) is defined as a Z-axis direction. Further, on a surface orthogonal to the Z-axis direction, an array direction of radiating elements is defined as an X axis and a direction orthogonal to the X axis is defined as a Y axis. Further, a positive direction of the Z axis in each drawing may be referred to as an upper side and a negative direction of the same may be referred to as a lower side.

(28) The dielectric substrate **130** is, for example, a low temperature co-fired ceramics (LTCC) multi-layer substrate; a multi-layer resin substrate, which is formed by laminating a plurality of resin layers made of resin such as epoxy or polyimide; a multi-layer resin substrate, which is formed by laminating a plurality of resin layers made of liquid crystal polymer (LCP) having a lower dielectric constant; a multi-layer resin substrate, which is formed by laminating a plurality of resin layers made of fluorine resin; a multi-layer resin substrate, which is formed by laminating a plurality of resin layers made of polyethylene terephthalate (PET) material; or a multi-layer substrate made of ceramics other than LTCC. Here, the dielectric substrate **130** does not necessarily have to have a multi-layer structure but may be a single-layer substrate.

(29) The dielectric substrate **130** has a rectangular shape in plan view in the normal direction (the Z-axis direction). The ground electrode GND is arranged over the entire surface of the dielectric substrate **130** at a position close to a lower surface **132** of the dielectric substrate **130**. The radiating

elements **122** and **123** are arranged adjacent to each other in the X-axis direction at positions close to an upper surface **131** (a surface in the positive direction of the Z axis) of the dielectric substrate **130**. The radiating elements **122** and **123** may be arranged to be exposed on the surface of the dielectric substrate **130** or may be arranged inside the dielectric substrate **130**.

(30) The radiating element **121** is arranged between the radiating element **122** and the ground electrode GND. In plan view in the normal direction of the dielectric substrate **130** (the Z-axis direction), the radiating element **122** overlaps with the radiating element **121**.

(31) Each of the radiating elements **121**, **122**, and **123** is a flat-plate like electrode having a rectangular shape. The size of the radiating element **122** is smaller than the size of the radiating element **121**, and the size of the radiating element **123** is further smaller than the size of the radiating element **122**. That is, the resonant frequency of the radiating element **122** is higher than the resonant frequency of the radiating element **121**, and the resonant frequency of the radiating element **123** is higher than the resonant frequency of the radiating element **122**. Accordingly, the frequency band (second frequency band) f2 of a radio wave radiated from the radiating element **122** is higher than the frequency band (first frequency band) f1 of a radio wave radiated from the radiating element **121**. Further, the frequency band (third frequency band) f3 of a radio wave radiated from the radiating element **123** is higher than the frequency band f2 of a radio wave radiated from the radiating element **122**. In the example of the first embodiment, center frequencies of the frequency bands of the radiating elements **121**, **122**, and **123** are 28 GHz, 39 GHz, and 60 GHz respectively.

(32) The radiating elements **121**, **122**, and **123** are supplied with a radio frequency signal from the RFIC **110** via the respective pieces of feed wiring **141**, **142**, and **143**. The feed wiring **141** is connected to a feed point SP1 of the radiating element **121** from the RFIC **110** through the ground electrode GND. Further, the feed wiring **142** is connected to a feed point SP2 of the radiating element **122** from the RFIC **110** through the ground electrode GND and the radiating element **121**. The feed wiring **143** is connected to a feed point SP3 of the radiating element **123** from the RFIC **110** through the ground electrode GND. The feed point SP1 is offset from the center of the radiating element **121** in the positive direction of the X axis. The feed point SP2 is offset from the center of the radiating element **122** in the positive direction of the X axis. The feed point SP3 is offset from the center of the radiating element **123** in the positive direction of the X axis. Accordingly, a radio wave whose polarization direction is the X-axis direction is radiated from each of the radiating elements **121**, **122**, and **123**.

(33) On the lower surface **132** of the dielectric substrate **130**, the RFIC **110** is mounted with solder bumps **160** interposed therebetween. However, the RFIC **110** may be connected to the dielectric substrate **130** with a multipole connector instead of solder connection.

(34) On the upper surface **131** of the dielectric substrate **130**, the dielectric layer **151** is arranged in a region (first region) RG1 covering the radiating elements **121** and **122**, and the dielectric layer **152** is arranged in a region (second region) RG2 covering the radiating element **123**. The dielectric constants of the dielectric layers **151** and **152** are both larger than the dielectric constant of the dielectric substrate **130**, and in addition, the dielectric constant ϵ_1 of the dielectric layer **151** is larger than the dielectric constant ϵ_2 of the dielectric layer **152** ($\epsilon_1 > \epsilon_2$). In the first embodiment, the thickness of the dielectric layer **151** is almost the same as the thickness of the dielectric layer **152**.

(35) Regarding a patch antenna having a flat-plate shape, assuming a Q value is lowered, a frequency bandwidth tends to expand, in general. A Q value is determined depending on a ratio of radiated power and stored power of a radiating element and a ground electrode. For example, assuming a distance between a radiating element and a ground electrode is increased or a dielectric constant between a radiating element and a ground electrode is lowered, a Q value decreases and a frequency bandwidth expands.

(36) Assuming an upper portion of a radiating element is covered by a dielectric layer having a higher dielectric constant than a dielectric substrate, a surface acoustic wave generated in the

radiating element tends to be stronger and lines of electric force generated in a direction from an end portion of the radiating element along an electrode surface travel farther compared to a configuration without a dielectric layer having a high dielectric constant. Thus, the path lengths of the lines of electric force from the radiating element to the ground electrode are elongated, which is consequently equivalent to a state in which the distance between the radiating element and the ground electrode is increased. Accordingly, by covering the upper portion of a radiating element by a dielectric layer having a high dielectric constant, the Q value of a patch antenna decreases and the frequency bandwidth consequently expands.

(37) The effect of a dielectric layer on surface acoustic waves increases as a dielectric constant of the dielectric layer increases. Therefore, the effect of expanding a frequency bandwidth increases as the dielectric constant is increased. The path length of lines of electric force, however, increases and therefore, unwanted mode resonance more easily occurs disadvantageously. That is, the expansion of the frequency bandwidth and the occurrence of unwanted mode resonance mutually have a trade-off relationship.

(38) Here, the effect of a dielectric layer on a surface acoustic wave tends to be sensitive as the frequency of a radio wave radiated from a radiating element increases. Therefore, assuming dielectric layers have the mutually-same thickness, it is necessary to lower the dielectric constant as the frequency of a radiated radio wave increases.

(39) Assuming three types of radiating elements in mutually-different frequency bands are arranged on a common dielectric substrate as the antenna module of this first embodiment, it is sometimes impossible to adapt the material and dimensions of the dielectric substrate to all the radiating elements due to restrictions in the area available for arrangement of the radiating elements on the dielectric substrate or manufacturing restrictions.

(40) For example, assuming the dielectric constant of a dielectric substrate is set to a dielectric constant adapted to a radiating element on the lower frequency side, the set dielectric constant may be too high for a radiating element on the higher frequency side. In this case, there is a possibility that a sufficient frequency bandwidth cannot be secured or unwanted mode resonance easily occurs due to a wavelength reduction effect. On the other hand, assuming the dielectric constant of a dielectric substrate is set to a dielectric constant adapted to a radiating element on the higher frequency side, the set dielectric constant is lower than a dielectric constant suitable for the thickness of the dielectric substrate with respect to the radiating element on the lower frequency side and therefore, the thickness of the dielectric substrate needs to be increased. This may cause inhibition in miniaturization of the antenna module.

(41) Assuming three types of radiating elements are arranged as the first embodiment, the radiating elements sometimes cannot be individually arranged due to the restriction of dimensions of a dielectric substrate. Especially in an array antenna in which a plurality of respective types of radiating elements are arrayed, the spacing between the radiating elements is too wide for the radiating elements on the higher frequency side, which may cause problems such as decrease in antenna gain and generation of grating lobes.

(42) Under the circumstances, in the antenna module **100** according to the first embodiment, two radiating elements **121** and **122** corresponding to mutually-close frequency bands are arranged in a stack structure among the three types of radiating elements, and the radiating elements **121** and **122** in the stack structure and the radiating element **123** are arranged adjacent to each other on the dielectric substrate **130**. Accordingly, an arrangement area of the radiating elements in the dielectric substrate **130** can be reduced compared to the configuration in which radiating elements are individually arranged.

(43) Further, in the antenna module **100**, a dielectric layer with a dielectric constant suitable for each radiating element is arranged in each of the region **RG1**, in which the radiating elements **121** and **122** in the stack structure are arranged, and the region **RG2**, in which the radiating element **123** is arranged, on the dielectric substrate **130**. This makes it possible to individually adjust the

strength of surface acoustic waves of the radiating elements on the lower frequency side (the radiating elements **121** and **122**) and the radiating element **123** on the higher frequency side. Accordingly, even assuming all the radiating elements are arranged in the common dielectric substrate **130**, the frequency bandwidths for the respective radiating elements can be appropriately expanded. Here, degradation in beam pattern caused by the surface acoustic wave of the radiating element **123** can be mitigated by setting the dielectric constant ϵ_1 of the dielectric layer **151** on the lower frequency side larger than the dielectric constant ϵ_2 of the dielectric layer **152** on the higher frequency side.

(44) Regarding the radiating elements **121** and **122** in the stack structure, the common dielectric layer **151** is provided for the mutually-different frequency bands. FIG. 3 shows the results of simulation regarding the effect of expanding the frequency bandwidths for the radiating elements **121** and **122** in such a configuration. FIG. 3 shows return loss in the case where the dielectric layer **151** is arranged (first embodiment) and the case where the dielectric layer **151** is not arranged (comparative example) in the region RG1 in which the radiating elements **121** and **122** are arranged. In FIG. 3, solid lines LN10 and LN15 indicate the return loss of the radiating element **121** and dashed lines LN11 and LN16 indicate the return loss of the radiating element **122**. Here, FIG. 3 evaluates bandwidths in which 6 dB return loss is achieved.

(45) As illustrated in FIG. 3, a bandwidth BW1L for the radiating element **121** of the first embodiment is larger than a bandwidth BW2L in the comparative example ($BW1L > BW2L$). Regarding a bandwidth ratio that is a ratio of a bandwidth with respect to a center frequency, the comparative example shows 27.1%, while the first embodiment shows 39.2%, which is improved.

(46) As for the radiating element **122**, a bandwidth BW1H in the first embodiment is larger than a bandwidth BW2H in the comparative example ($BW1H > BW2H$), and the bandwidth ratio is also improved from 12.2% to 17.6%.

(47) Thus, even assuming radiating elements are arranged in a stack structure, the frequency bands for both of the radiating elements **121** and **122** can be expanded by arranging a dielectric layer having a higher dielectric constant than the dielectric substrate on the upper surface of the dielectric substrate.

(48) As for the radiating element **123** which is individually arranged, the frequency band can be expanded by arranging the dielectric layer **152**, which has the dielectric constant suitable for the radiating element **123**, compared to the configuration without the dielectric layer **152**.

(49) As described above, in an antenna module in which radiating elements corresponding to three mutually-different frequency bands are arranged, two radiating elements corresponding to mutually-close frequency bands are arranged as a stack structure in a manner to be adjacent to the rest of the radiating elements, and dielectric layers having dielectric constants, which correspond to the radiating elements to be arranged, are arranged in respective regions, in which the radiating elements are arranged, on the dielectric substrate. This configuration can suppress increase in an arrangement area in the dielectric substrate and realize broadband in antenna characteristics of each of the radiating elements.

(50) Here, the “radiating elements **121**, **122**, and **123**” of the first embodiment correspond to a “first radiating element”, a “second radiating element”, and a “third radiating element” in the present disclosure respectively. The “upper surface **131**” and the “lower surface **132**” of the dielectric substrate **130** in the first embodiment correspond to a “first surface” and a “second surface” in the present disclosure respectively.

First Modification

(51) The first embodiment has described the configuration in which two radiating elements which are on the relatively lower frequency side are formed in a stack structure among three radiating elements. A first modification will describe a configuration in which two radiating elements which are on the relatively higher frequency side are formed in a stack structure among three radiating elements.

(52) FIG. 4 is a side perspective view of an antenna module **100A** according to a first modification. Referring to FIG. 4, in an antenna device **120A** of the antenna module **100A**, the radiating element **121** corresponding to the lowest frequency band is singularly arranged on the dielectric substrate **130** and two radiating elements **122** and **123** on the relatively higher frequency side are arranged as a stack structure in the dielectric substrate **130**, among the three radiating elements. Specifically, the radiating element **122** is arranged between the radiating element **123** and the ground electrode GND in the normal direction of the dielectric substrate **130** (the Z-axis direction). Not illustrated in FIG. 4, the radiating element **123** overlaps with the radiating element **122**, in plan view in the normal direction of the dielectric substrate **130**. The feed wiring **143** for transmitting a radio frequency signal to the radiating element **123** is connected to the feed point SP3 of the radiating element **123** from the RFIC **110** through the ground electrode GND and the radiating element **122**.

(53) On the dielectric substrate **130**, the dielectric layer **151** is arranged in the region RG1 covering the radiating element **121**, and the dielectric layer **152** is arranged in the region RG2 covering the radiating elements **122** and **123**. Here, the dielectric constants of the dielectric layers **151** and **152** do not necessarily have to be the same as the dielectric constants of the first embodiment and the dielectric constants are appropriately selected depending on characteristics required for each radiating element.

(54) The configuration of the antenna module **100A** of the first modification is preferable assuming the frequency band for the radiating element **122** is closer to the frequency band for the radiating element **123** than the frequency band for the radiating element **121**.

(55) As described above, in the antenna module **100A** according to the first modification as well, two radiating elements corresponding to mutually-close frequency bands are arranged as a stack structure, and dielectric layers having dielectric constants, which correspond to the radiating elements to be arranged, are arranged in respective regions, in which the radiating elements are arranged, on the dielectric substrate. This configuration can suppress increase in an arrangement area in the dielectric substrate and realize broadband in antenna characteristics of each of the radiating elements.

Second Modification

(56) A second modification will describe an antenna module having four types of radiating elements corresponding to mutually-different frequency bands.

(57) FIG. 5 is a side perspective view of an antenna module **100B** according to a second modification. Referring to FIG. 5, an antenna device **120B** of the antenna module **100B** includes a radiating element **124** in addition to the radiating elements **121**, **122**, and **123**. In the example of the antenna module **100B**, a frequency band (fourth frequency band) f_4 of a radio wave radiated from the radiating element **124** is lower than the frequency bands of the radio waves radiated from the radiating elements **121**, **122**, and **123** ($f_4 < f_1 < f_2 < f_3$).

(58) In the antenna module **100B**, the radiating elements **121** and **124** are arranged in a stack structure and the radiating elements **122** and **123** are arranged in a stack structure. The radiating element **124** is arranged between the radiating element **121** and the ground electrode GND. The radiating element **124** is supplied with a radio frequency signal from the RFIC **110** via feed wiring **144**. The feed wiring **144** is connected to a feed point SP4 of the radiating element **124** from the RFIC **110** through the ground electrode GND. The feed wiring **141** for supplying a radio frequency signal to the radiating element **121** is connected to the feed point SP1 of the radiating element **121** from the RFIC **110** through the ground electrode GND and the radiating element **124**.

(59) On the dielectric substrate **130**, the dielectric layer **151** is arranged in the region RG1 covering the radiating elements **121** and **124**, and the dielectric layer **152** is arranged in the region RG2 covering the radiating elements **122** and **123**. The dielectric constants of the dielectric layers **151** and **152** are determined depending on characteristics required for each radiating element.

(60) The above description is provided on the example in which the frequency band of the radio wave radiated from the radiating element **124** is lower than those of other radiating elements **121**,

122, and **123**, but the frequency band for the radiating element **124** is not limited to this.

Combinations of radiating elements forming a stack structure are appropriately determined depending on the frequency band of the radiating element **124**.

(61) As described above, in the antenna module including four types of radiating elements corresponding to mutually-different frequency bands as well, two radiating elements corresponding to mutually-close frequency bands are arranged as a stack structure, and dielectric layers having dielectric constants, which correspond to the radiating elements to be arranged, are arranged in respective regions, in which the radiating elements are arranged, on the dielectric substrate. This configuration can suppress increase in an arrangement area in the dielectric substrate and realize broadband in antenna characteristics of each of the radiating elements.

(62) The “radiating element **124**” in the second modification corresponds to a “fourth radiating element” in the present disclosure.

Second Embodiment

(63) A second embodiment will describe an example in which an antenna module is an array antenna in which a plurality of electrodes included in respective radiating elements are arranged in an array.

(64) FIG. **6** is a plan view of an antenna module **100C** according to the second embodiment. In an antenna device **120C** of the antenna module **100C** in FIG. **6**, electrodes included in respective radiating elements are alternately arranged in the X-axis direction and the Y-axis direction to form an array. More specifically, the electrodes of the radiating elements **122** and **123** on the higher frequency side form a stack structure, and the electrode of the radiating element **121** is singularly arranged on the dielectric substrate **130**, as the above-described first modification. The radiating elements **121** and the stack structures of the radiating elements **122** and **123** are alternately arranged in the X-axis direction and the Y-axis direction.

(65) On the dielectric substrate **130**, the dielectric layer **151** is arranged in the region RG1 covering the electrode of the radiating element **121**, and the dielectric layer **152** is arranged in the region RG2 covering the electrodes of the radiating elements **122** and **123**. Here, plan views of FIG. **6** and later-described FIGS. **11** and **12** omit hatching for a portion overlapping with each radiating element in the dielectric layers **151** and **152**, for the sake of simpler description.

(66) As described above, in the array antenna as well, two radiating elements corresponding to mutually-close frequency bands are arranged as a stack structure among the three types of radiating elements corresponding to mutually-different frequency bands, and dielectric layers having dielectric constants, which correspond to the radiating elements to be arranged, are arranged in respective regions, in which the radiating elements are arranged, on the dielectric substrate. This configuration can suppress increase in an arrangement area in the dielectric substrate and realize broadband in antenna characteristics of each of the radiating elements.

Third Modification

(67) A third modification will describe an example of a configuration in which dielectric layers arranged in respective regions have mutually different thickness.

(68) FIG. **7** is a side perspective view of an antenna module **100D** according to the third modification. An antenna device **120D** of the antenna module **100D** is an array antenna in which the electrodes of the radiating elements **122** and **123** on the higher frequency side are provided in a stack structure, and the electrode of the radiating element **121** on the lower frequency side is singularly arranged, as is the case with the second embodiment. Here, FIG. **7** illustrates an example of the configuration in which each of the radiating elements **121**, **122**, and **123** includes two electrodes.

(69) The dielectric layer **151** is arranged in the region RG1 covering the electrodes of the radiating element **121**, and the dielectric layer **152** is arranged in the region RG2 covering the electrodes of the radiating elements **122** and **123** forming the stack structure. Here, the thickness (dimension in the Z-axis direction) D1 of the dielectric layer **151** is larger than the thickness D2 of the dielectric

layer **152** ($D1 > D2$) in the antenna module **100D**. Assuming the thickness of a dielectric layer is increased, a substantial dielectric constant viewed from a radiating element increases. Accordingly, the configuration as that of the antenna module **100D** can make the dielectric constant of the dielectric layer **151** further larger than the dielectric constant of the dielectric layer **152**, compared to the configuration in which the dielectric layers have the same thickness as the first embodiment. Further, even assuming the dielectric layer **151** and the dielectric layer **152** are made of the same materials, the substantial dielectric constant of the dielectric layer **151** viewed from the radiating element can be increased to be larger than that of the dielectric layer **152** by allowing the dielectric layer **151** to have the larger thickness than the thickness of the dielectric layer **152**.

Fourth Modification

(70) A fourth modification will describe a configuration in which the position of a ground electrode in a dielectric substrate varies depending on regions in which dielectric layers are arranged.

(71) FIG. **8** is a side perspective view of an antenna module **100E** according to the fourth modification. An antenna device **120E** of the antenna module **100E** is an array antenna in which the electrodes of the radiating elements **122** and **123** on the higher frequency side are provided in a stack structure, and the electrode of the radiating element **121** on the lower frequency side is singularly arranged, as is the case with the third modification. The thickness of the dielectric layer **151** and the thickness of the dielectric layer **152** are set to be the same as each other in the antenna module **100E**. Further, in the antenna module **100E**, the ground electrode GND in the region RG1 in which the dielectric layer **151** is arranged is arranged on a position closer to the upper surface **131** than the ground electrode GND in the region RG2 in which the dielectric layer **152** is arranged. In other words, a distance H1 between the radiating element **121** and the ground electrode GND is shorter than a distance H2 between the radiating element **123** and the ground electrode GND.

(72) In general, the frequency bandwidth of a radiating element in a patch antenna is affected by the distance between the radiating element and a ground electrode, and the frequency bandwidth is increased as the distance increases. In the stack structure of the radiating elements in the region RG2, the radiating element **123** is coupled with the radiating element **122** to operate as an antenna, and the radiating element **122** substantially functions as a ground electrode. Meanwhile, the radiating element **122** is coupled with the ground electrode GND to operate as an antenna. Accordingly, the overall thickness of the dielectric substrate **130** (that is, the distance H2 between the radiating element **123** and the ground electrode GND) is sometimes increased so as to ensure desired frequency bandwidth for the radiating elements **122** and **123**. In this case, the distance to the ground electrode GND from the radiating element **121**, which is singularly arranged in the dielectric substrate **130**, may be longer than the distance suitable for characteristics of the radiating element **121**.

(73) In the antenna module **100E**, the distance between the radiating element **121** and the ground electrode GND can be set to the distance suitable for the characteristics of the radiating element **121** by setting the position of the ground electrode GND in the region RG1, in which the radiating element **121** on the lower frequency side is arranged, closer to the upper surface **131** than the position of the ground electrode GND in the region RG2. Accordingly, degradation in antenna characteristics in each radiating element can be suppressed.

Fifth Modification

(74) The fourth modification has described the configuration in which the distance between the radiating element **121** and the ground electrode GND is adjusted by the position of the ground electrode GND. A fifth modification will describe a configuration in which the ground electrode GND is arranged at the same position in the region RG1 and the region RG2 and the distance from the ground electrode GND is adjusted by the position of the radiating element **121**.

(75) FIG. **9** is a side perspective view of an antenna module **100F** according to the fifth modification. In an antenna device **120F** of the antenna module **100F**, the radiating element **121**, which is singularly arranged in the dielectric substrate **130**, is arranged closer to the ground

electrode GND than the radiating element **123**. This configuration can set the distance H1 between the radiating element **121** and the ground electrode GND to the distance suitable for the antenna characteristics of the radiating element **121**.

(76) Here, there is part of the dielectric substrate **130** on the closer side to the upper surface **131** than the radiating element **121** in the antenna module **100F**. Therefore, the total amount of dielectric in the radiation direction is larger than that in the antenna module **100E** of the fourth modification. Accordingly, the dielectric constant of the dielectric layer **151** arranged in the region RG1 is preferably adjusted depending on the amount of the dielectric substrate **130** between the radiating element **121** and the upper surface **131**.

Sixth Modification

(77) A sixth modification will describe a configuration in which the ground electrode GND is arranged at the same position like the fifth modification and the thickness of dielectric layers on radiating elements differs in the region RG1 and the region RG2.

(78) FIG. **10** is a side perspective view of an antenna module **100G** according to the sixth modification. In an antenna device **120G** of the antenna module **100G**, the ground electrode GND is arranged on the same position in the Z-axis direction over the entire area of the dielectric substrate **130**, and the position of the radiating element **121**, which is arranged in the region RG1, is closer to the ground electrode GND than the radiating element **123**, as is the case with the fifth modification. Further, the dielectric layer **151** in the region RG1 is arranged up to the position at which the dielectric layer **151** is in contact with the radiating element **121** in the antenna module **100G**. In other words, the upper surface **131** on the dielectric substrate **130** in the antenna module **100G** has a step between the region RG1 and the region RG2, and the thickness (the dimension in the Z-axis direction) of the dielectric substrate **130** in the region RG1 is smaller than the thickness of the dielectric substrate **130** in the region RG2. In the region RG1, the dielectric layer **151** is formed thicker than the dielectric layer **152** so as to fill the above-mentioned smaller thickness.

(79) This configuration can eliminate boundaries of the dielectric layers in the radiation direction of the radiating element **121**, being able to reduce return loss caused by boundaries of dielectric layers. Further, the dielectric constant of the dielectric layer on the radiating element **121** can be further increased, being able to realize broadband in the antenna characteristics of the radiating element **121**.

Seventh Modification

(80) The second embodiment and the third modification to the sixth modification have described the configuration in which the radiating elements, which are singularly arranged, and the radiating elements, which form the stack structures, are alternately arranged on the dielectric substrate. A seventh modification will describe a configuration in which radiating elements, which are singularly arranged, and radiating elements, which form the stack structures, are collectively arranged in individual regions on a dielectric substrate.

(81) FIG. **11** is a plan view of an antenna module **100H** according to the seventh modification. In an antenna device **120H** of the antenna module **100H**, six pieces of radiating elements **121** are two-dimensionally arrayed in the region RG1 in the negative direction of the X axis in the dielectric substrate **130**, and six pairs of stack structures of the radiating elements **122** and **123** are two-dimensionally arrayed in the region RG2 in the positive direction of the X axis in the dielectric substrate **130**. The radiating elements **121** are covered by the dielectric layer **151** in the region RG1, and the radiating elements **122** and **123** are covered by the dielectric layer **152** in the region RG2.

(82) In the configuration in which radiating elements are alternately arranged as the second embodiment, an inter-element pitch between radiating elements corresponding to the same frequency is set to a pitch corresponding to the radiating element **121** on the lower frequency side. In this configuration, the pitch of the radiating elements **122** and **123** on the higher frequency side may be excessively larger than a pitch suitable for these radiating elements. As a result, the antenna

characteristics may be degraded due to reduction in antenna gain or generation of grating lobes.

(83) By collectively arranging the radiating elements **121**, which are on the lower frequency side, in the region **RG1** and collectively arranging the radiating elements **122** and **123**, which are on the higher frequency side, in the region **RG2** as the seventh modification, the pitch between the radiating elements on the higher frequency side on which the inter-element pitch is relatively small can be set smaller than the pitch of the radiating elements on the lower frequency side, being able to suppress degradation in the antenna characteristics caused by restriction of the inter-element pitch.

Third Embodiment

(84) The above-described embodiments and modifications have described the configuration in which each radiating element radiates a radio wave in one polarization direction. A third embodiment will describe a configuration in which features of the present disclosure are applied to a so-called dual polarization type antenna module that can radiate radio waves in two different polarization directions.

(85) FIG. **12** is a plan view of an antenna module **100I** according to the third embodiment. Referring to FIG. **12**, an antenna device **120I** of the antenna module **100I** is an array antenna in which the radiating elements **121** and the stack structures of the radiating elements **122** and **123** are alternately arranged in a manner to be adjacent to each other, as is the case with the antenna module **100C** of FIG. **6**. In the antenna module **100I**, two feed points are provided with respect to each of the radiating elements **121**, **122**, and **123**.

(86) More specifically, a feed point **SP11** is arranged on a position offset from the center of the electrode in the positive direction of the X axis and a feed point **SP12** is arranged on a position offset from the center of the electrode in the negative direction of the Y axis, in the radiating element **121**. Assuming a radio frequency signal is supplied to the feed point **SP11**, a radio wave whose polarization direction is the X-axis direction is radiated from the radiating element **121**. Meanwhile, assuming a radio frequency signal is supplied to the feed point **SP12**, a radio wave whose polarization direction is the Y-axis direction is radiated from the radiating element **121**.

(87) Similarly, a feed point **SP21** is arranged on a position offset from the center of the electrode in the positive direction of the X axis and a feed point **SP22** is arranged on a position offset from the center of the electrode in the negative direction of the Y axis, in the radiating element **122**. Assuming a radio frequency signal is supplied to the feed point **SP21**, a radio wave whose polarization direction is the X-axis direction is radiated from the radiating element **122**.

Meanwhile, assuming a radio frequency signal is supplied to the feed point **SP22**, a radio wave whose polarization direction is the Y-axis direction is radiated from the radiating element **122**.

(88) A feed point **SP31** is arranged on a position offset from the center of the electrode in the negative direction of the X axis and a feed point **SP32** is arranged on a position offset from the center of the electrode in the positive direction of the Y axis, in the radiating element **123**.

Assuming a radio frequency signal is supplied to the feed point **SP31**, a radio wave whose polarization direction is the X-axis direction is radiated from the radiating element **123**.

Meanwhile, assuming a radio frequency signal is supplied to the feed point **SP32**, a radio wave whose polarization direction is the Y-axis direction is radiated from the radiating element **123**.

(89) The radiating element **122** and the radiating element **123** overlap with each other in plan view in the normal direction of the dielectric substrate **130**. Therefore, assuming the positions of feed points corresponding to the same polarization are set in the same direction in the radiating element **122** and the radiating element **123**, pieces of feed wiring may come close to be coupled to each other, which may cause degradation in isolation. The feed points of the radiating element **122** are arranged on positions in the opposite direction to positions of the feed points of the radiating element **123**, which can suppress degradation in isolation characteristics between radio waves radiated from the radiating element **122** and radio waves radiated from the radiating element **123**.

(90) In the above-described antenna module **100I** of the dual polarization type as well, the

dielectric layer **151** is arranged in the region RG1, in which the radiating element **121** is arranged, and the dielectric layer **152** is arranged in the region RG2, in which the radiating elements **122** and **123** are arranged, being able to realize broadband in antenna characteristics of each radiating element.

(91) The “X-axis direction” and the “Y-axis direction” in the third embodiment correspond to a “first direction” and a “second direction” in the present disclosure respectively.

(92) The above-described embodiments and modifications have described the example in which the radiating element is a patch antenna having a flat-plate shape. However, the above-mentioned features are applicable to a configuration in which each radiating element is a loop antenna.

(93) The configuration in which the dielectric layers **151** and **152** are arranged to be in contact with the dielectric substrate **130** has been described for each of the above-mentioned antenna modules. However, a configuration may be employed in which the dielectric layers **151** and **152** are arranged to be separate from the dielectric substrate **130** and an air layer is provided between the dielectric layers **151** and **152** and the dielectric substrate **130**. For example, a configuration may be employed in which the dielectric layers **151** and **152** are arranged on a casing of the communication device **10** and each radiating element of the dielectric substrate **130** arranged on a mounting substrate is arranged to face the dielectric layers **151** and **152**.

(94) Further, the configuration in which each radiating element and the ground electrode GND are arranged in the same dielectric substrate **130** has been described for each of the above-mentioned antenna modules. However, the radiating element may be arranged in a substrate different from the substrate in which the ground electrode GND is arranged. In this configuration, pieces of feed wiring are connected by solder bumps or other connecting members between the two substrates.

(95) Furthermore, in the regions RG1 and RG2, the upper surfaces of the dielectric layers **151** and **152** are flat, but the upper surfaces of the dielectric layers **151** and **152** may be uneven.

(96) The embodiments disclosed here should be considered exemplary and not restrictive in all respects. The scope of the present disclosure is indicated by the scope of the claims rather than the description of the above-described embodiments, and is intended to include all changes within the scope and meaning equivalent to the scope of the claims.

REFERENCE SIGNS LIST

(97) **10** communication device **100**, **100A** to **100I** antenna module **110** RFIC **110A** to **110C** feed circuit **111A** to **111D**, **113A** to **113D**, **117** switch **112AR** to **112DR** low noise amplifier **112AT** to **112DT** power amplifier **114A** to **114D** attenuator **115A** to **115D** phase shifter **116** signal synthesizer/demultiplexer **118** mixer **119** amplifying circuit **120**, **120A** to **120I** antenna device **121** to **124** radiating element **130** dielectric substrate **141** to **144** feed wiring **151**, **152** dielectric layer **160** solder bump **200** BBIC GND ground electrode RG1, RG2 region SP1 to SP4, SP11, SP12, SP21, SP22, SP31, SP32 feed point

Claims

1. An antenna module comprising: a dielectric substrate that has a first surface and a second surface; a first radiating element, a second radiating element, and a third radiating element that are arranged in the dielectric substrate, have a flat-plate shape, and can radiate respective radio waves in mutually-different frequency bands; a first dielectric layer that is arranged on the first surface in a manner to cover a first region in which the first radiating element is arranged; and a second dielectric layer that is arranged on the first surface in a manner to cover a second region in which the third radiating element is arranged, wherein the first region and the second region are adjacent to each other, a dielectric constant of the first dielectric layer and a dielectric constant of the second dielectric layer are higher than a dielectric constant of the dielectric substrate, the first radiating element can radiate a radio wave in a first frequency band, the second radiating element can radiate a radio wave in a second frequency band, the second frequency band being higher than the first

frequency band, the third radiating element can radiate a radio wave in a third frequency band, the third frequency band being higher than the second frequency band, and the second radiating element overlaps with the first radiating element or the third radiating element in plan view in a normal direction of the dielectric substrate.

2. The antenna module according to claim 1, further comprising: a ground electrode that is arranged in a manner to be opposed to the first radiating element, the second radiating element, and the third radiating element, wherein the second radiating element overlaps with the first radiating element, and the first radiating element is arranged between the second radiating element and the ground electrode.

3. The antenna module according to claim 1, further comprising: a ground electrode that is arranged in a manner to be opposed to the first radiating element, the second radiating element, and the third radiating element, wherein the second radiating element overlaps with the third radiating element, and the second radiating element is arranged between the third radiating element and the ground electrode.

4. The antenna module according to claim 3, wherein a distance between the first radiating element and the ground electrode is shorter than a distance between the third radiating element and the ground electrode.

5. The antenna module according to claim 4, wherein the ground electrode in the first region is arranged closer to the first surface than the ground electrode in the second region.

6. The antenna module according to claim 4, wherein the first radiating element is arranged closer to the ground electrode than the third radiating element.

7. The antenna module according to claim 6, further comprising: a fourth radiating element that can radiate a radio wave in a fourth frequency band, the fourth frequency band being lower than the first frequency band, wherein the fourth radiating element overlaps with the first radiating element in plan view in the normal direction of the dielectric substrate.

8. The antenna module according to claim 7, wherein the dielectric constant of the first dielectric layer is higher than the dielectric constant of the second dielectric layer.

9. The antenna module according to claim 8, wherein the first dielectric layer is thicker than the second dielectric layer.

10. The antenna module according to claim 9, wherein each of the first radiating element, the second radiating element, and the third radiating element is configured to be able to radiate radio waves in two different polarization directions.

11. The antenna module according to claim 10, wherein a radio frequency signal is supplied to a first feed point and a second feed point in each of the first radiating element, the second radiating element, and the third radiating element, the first feed point is arranged on a position offset in a first direction from a center of a corresponding radiating element, and the second feed point is arranged on a position offset in a second direction from a center of a corresponding radiating element, the second direction being different from the first direction.

12. An antenna module comprising: a dielectric substrate that has a first surface and a second surface; a first radiating element, a second radiating element, and a third radiating element that are arranged in the dielectric substrate and each of which includes a plurality of electrodes having a flat-plate shape; a first dielectric layer that is arranged on the first surface in a manner to cover a first region in which the electrodes of the first radiating element are arranged; and a second dielectric layer that is arranged on the first surface in a manner to cover a second region in which the electrodes of the third radiating element are arranged, wherein the first radiating element, the second radiating element, and the third radiating element can radiate radio waves in mutually-different frequency bands, the first region and the second region are adjacent to each other, a dielectric constant of the first dielectric layer and a dielectric constant of the second dielectric layer are higher than a dielectric constant of the dielectric substrate, the first radiating element can radiate a radio wave in a first frequency band, the second radiating element can radiate a radio

wave in a second frequency band, the second frequency band being higher than the first frequency band, the third radiating element can radiate a radio wave in a third frequency band, the third frequency band being higher than the second frequency band, and the electrodes of the second radiating element overlap with the electrodes of the third radiating element in plan view in a normal direction of the dielectric substrate.

13. The antenna module according to claim 12, further comprising: a feed device that supplies a radio frequency signal to each radiating element.

14. A communication device on which the antenna module according to claim 13 is mounted.

15. The antenna module according to claim 3, further comprising: a fourth radiating element that can radiate a radio wave in a fourth frequency band, the fourth frequency band being lower than the first frequency band, wherein the fourth radiating element overlaps with the first radiating element in plan view in the normal direction of the dielectric substrate.

16. The antenna module according to claim 1, wherein the dielectric constant of the first dielectric layer is higher than the dielectric constant of the second dielectric layer.

17. The antenna module according to claim 1, wherein the first dielectric layer is thicker than the second dielectric layer.

18. The antenna module according to claim 1, wherein each of the first radiating element, the second radiating element, and the third radiating element is configured to be able to radiate radio waves in two different polarization directions.

19. The antenna module according to claim 1, wherein a radio frequency signal is supplied to a first feed point and a second feed point in each of the first radiating element, the second radiating element, and the third radiating element, the first feed point is arranged on a position offset in a first direction from a center of a corresponding radiating element, and the second feed point is arranged on a position offset in a second direction from a center of a corresponding radiating element, the second direction being different from the first direction.

20. The antenna module according to claim 1, further comprising: a feed device that supplies a radio frequency signal to each radiating element.
